

ANKIT G. MISHRA

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EDUCATION

University of Mumbai Bachelor of Technology in Artificial Intelligence & Machine Learning GPA: 3.5 / 4	Mumbai, India 2022- 2025
MSBTE Diploma in Computer Science Engineering Percentage: 87%	Thane, India 2019 – 2022

WORK EXPERIENCE

DocuraHealth (YC W26) <i>Founding AI Engineer</i>	San Francisco, USA Jan 2026 – June 2026
● Built and improved a Planning Agent system for report generation workflows, orchestrating extraction planning, sample coordination, report preparation, and iterative refinement/replanning for failed or low-confidence outputs, reducing manual operational effort by ~50%. ● Developed Verification Agent layers across multiple AI pipelines to validate extracted data and generated reports, reducing report inconsistencies and manual review/correction workload by ~40% for operations teams. ● Worked on Context Handling systems to improve reasoning consistency and structured output reliability, leading to noticeably higher report quality and reduced regeneration cycles. ● Optimized report generation and Review of Records (ROR) workflows using asynchronous processing and parallel execution strategies, decreasing processing latency by ~60% and significantly improving throughput for large-scale healthcare record handling. ● Enhanced OCR and document extraction pipelines by improving prompt behavior, extraction orchestration, and validation logic, improving extraction accuracy and reducing downstream correction effort. ● Integrated and onboarded new healthcare client workflows, ensuring generated outputs matched client-specific templates, formatting standards, and reporting requirements while reducing template adjustment/review effort by ~50%. ● Worked on Phone Agent workflows capable of processing call/audio inputs and converting extracted information into structured healthcare forms, reducing manual form-filling effort for operational workflows. ● Reverse engineered a proprietary .ds2 (Olympus CELP-based) audio format and built a custom decoding pipeline (DS2 → WAV → OGG Opus) to enable accurate transcription using Deepgram/Whisper, overcoming lack of native FFmpeg support ● Led end-to-end development of AI infrastructure, workflow automation systems, frontend upload/review flows, and production healthcare AI pipelines as the primary engineer. ● Tech Stack: Python, Next.js, FastAPI, Docker, LlamaIndex, Deepgram, Whisper, AWS, AWS Textract, DataLab OCR, Multi-Agent Systems, Voice AI Agents, Prompt Engineering, AI Verification Systems.	
Pibit.ai (YC W21) Series-A <i>Machine Learning Engineer</i>	Bangalore, India June 2025 – Dec 2025
● Automated Workflows: Developed a Python-based comparison script leveraging pandas and openpyxl to automate validation of our product between ground truth and model output, enabling the Quality Assurance and Operations teams to reduce manual review time by ~70%. The script flags discrepancies and critical issues in red, producing instantly actionable reports. ● Prompt Engineering for Extraction Accuracy: Significantly enhanced the performance of the Exmod LLM system by refining prompts and testing response behavior using Langfuse, leading to a ~25% improvement in document-extraction accuracy and reliability. ● Resolved Data Contamination issues: Identified and fixed contamination and misalignment issues in various Documents, improving data integrity and downstream extraction performance. ● Optimized Supplemental Document Processing: Improved prompt reliability and backend processing logic, increasing the extraction precision of supplemental documents and saving the team ~40% of manual QA and correction time.	

- Collaborated with cross-functional engineering teams to develop the *Piparse* package, standardizing shared document ingestion, preprocessing, and extraction workflows across multiple client pipeline
- Contributing to the design and development of a **prompt management system to centralize prompt versioning, experimentation for large-scale LLM workflows.**
- **Tech Stack:** Python, Langchain, LangGraph, Pandas, FastAPI, Openpyxl, Prompt Engineering, Langfuse, AWS S3, AWS CloudWatch

AAROGYA ID

Founding AI Engineer

Hyderabad, India

Jan 2025 – May 2025

- Designed and evaluated a medical question-answering system by testing and benchmarking multiple large language models (Claude Sonnet 3.5, OpenBioLLM, Meditron), selecting the most accurate and contextually relevant output patterns for production use.
- **Prompt Engineering for Document Digitization:** Engineered and optimized prompts for AWS Textract and Claude 3.5, leading to a **~40% reduction in extraction errors** during medical document digitization workflows.
- **LLM Integration at Scale:** Integrated **Gemini Flash 2.0** into the backend digitization pipeline, significantly improving parsing performance and boosting throughput by **~60%** on large PDF files.
- **End-to-End Backend Integration:** Led the full backend integration of Gemini Flash 2, handling complex API responses using **AWS Lambda, Bedrock, and CloudWatch** to monitor and scale system behavior.
- **Cost Efficient Infrastructure Refactoring:** Migrated backend storage from **AWS RDS to DynamoDB**, achieving a **notable reduction in storage costs** while maintaining query performance.
- **Secure API Deployment:** Designed and deployed RESTful APIs via **AWS API Gateway**, ensuring scalable and secure access to digitization services.
- **Technical Documentation & Evaluation Reports:** Authored comprehensive documentation for the system's architecture, LLM evaluation protocols, API structure, and workflow automation to ensure smooth handoff and maintainability.
- **Tech Stack:** Claude Sonnet 3.5, OpenBioLLM, Meditron, Gemini Flash 2, AWS Lambda, AWS Bedrock, AWS CloudWatch, AWS API Gateway, AWS Textract, AWS RDS, DynamoDB, FastAPI, React.js, Prompt Engineering, REST APIs,

ADDITIONAL EXPERIENCE (Internships / Part time)

Machine Learning Engineer | Software Engineer | Python Instructor

India

Nov 2022 – Aug 2024

- Worked across **5 startups + 1 open-source organization** (part-time/intern): Verbient Technologies, Shoziki Infotech, HomeZing, TCET Open Source, Homestogether, and Universe Simplified.
- Independently built and deployed CMS platforms and full-stack web applications, managing frontend, backend, deployment, and SEO optimization across multiple startup projects.
- Led end-to-end feature development and technical execution for client and internal projects, taking ownership from development to deployment in fast-paced startup environments.
- Developed Statistical ML-powered tools, recommendation systems, and automation-focused solutions to improve personalization, workflow efficiency, and data-driven decision making.
- Worked on RAG-based chatbot systems and AI-powered product features using LLM workflows to enhance user engagement, contextual search, and information retrieval pipelines.
- Contributed to backend engineering, API integrations, client-side validation systems, and business logic implementation for scalable web applications.
- Deployed and managed applications on AWS EC2 environments while handling production-level debugging, updates, and optimization workflows.
- Collaborated with cross-functional teams, founders, and developers while also taking independent ownership of critical engineering tasks and deliverables.
- Mentored and taught Python programming at Universe Simplified, helping university students and teaching professionals understand programming fundamentals and practical software development concepts.
- **Tech Stack:** Langchain, GPT 4.1, Scikit Learn, NLTK, BeautifulSoup4, Selenium, Python, Flask, Django, React, PHP, Javascript, SQL, Xampp, Git, Github, AWS Ec2 Windows,

PROJECTS

Projects :

India, Nov 2022 - *Present*

- **Mutant-AI Visit:** [mutant-ai](#)

Developed an open-source Python library for **LLM evaluation, dataset generation, and red teaming** for AI agents and RAG systems.

Mutate Engine: Automatically expands existing evaluation datasets by generating behavioral variations across multiple dimensions (Emotion, Angry, Frustrated , Prompt Injection, Social Engineering, Context-Manipulation, etc.), enabling broader and more comprehensive evaluation coverage.

Red Team Engine: Executes automated adversarial attacks against LLMs and AI agents—including prompt injection, jailbreak, instruction-conflict, context-poisoning, and other attack strategies—to identify vulnerabilities, evaluate security posture, and improve application robustness.

- [Visit My Website For More Projects - Project Section](#)

TECHNICAL SKILLS

Frameworks: HuggingFace, LangGraph, Langchain, LlamaIndex, Tensorflow, Scikit Learn, PyTorch, Keras, Numpy, Pandas, NLTK, Spacy, Flask, FastAPI, Crew AI, PhiData, PydanticAI, Next.js, Vapi

Tools: Git, Github Actions, ArgoCD, HELM, DVC, MLflow, Postman, DataStax, Unsloth, pgAdmin4, Claude Code

Databases: MySQL, MongoDB (Familiar), VectorDatabases

Infrastructure & Cloud: Langfuse, Langsmith, Docker, Kubernetes, AWS EKS, AWS Bedrock, AWS Lambda, AWS S3 Bucket, AWS Ec2, AWS RDS, AWS CloudWatch